Evaluating AI Agents

Decomposing Agents





DeepLearning.AI

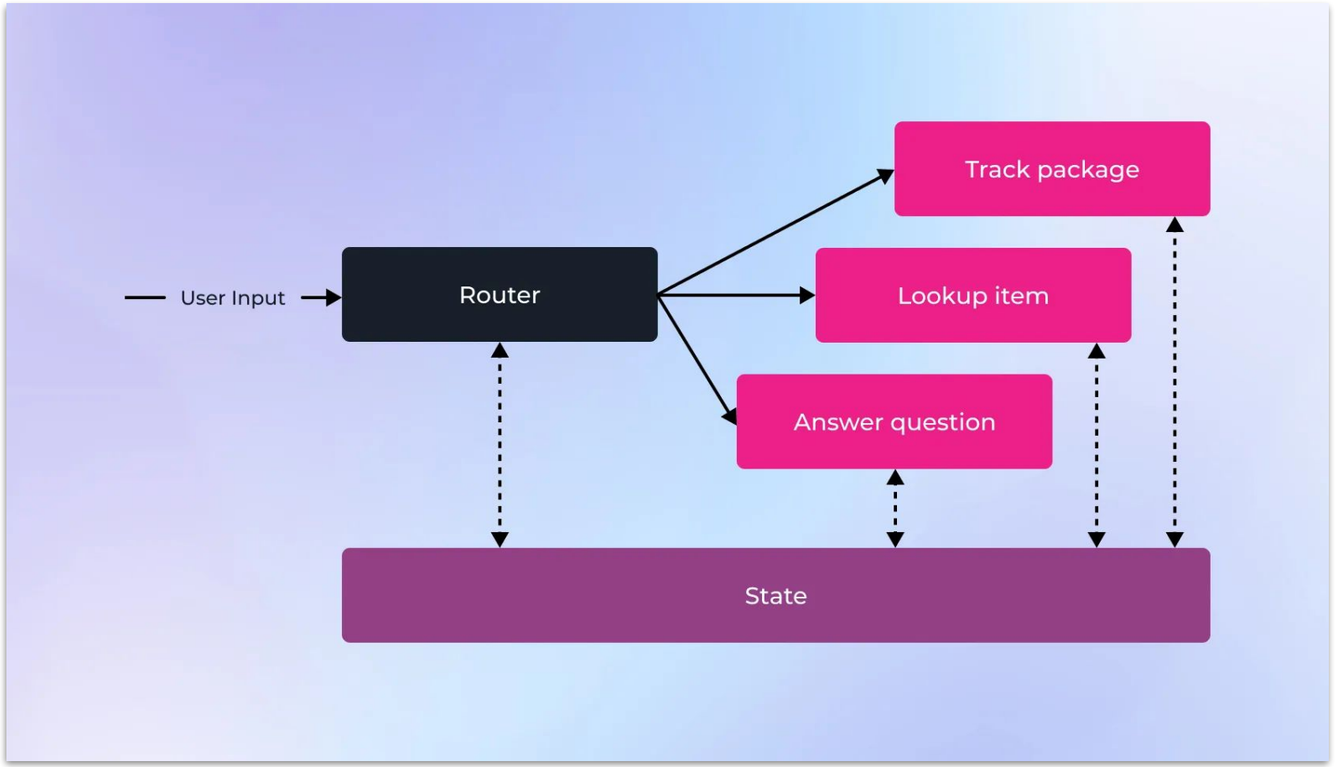
Talking Head

In this lesson:

- Understanding the basic structure of agents
- Examining an example agent

The Main Components

1. A Router
2. Skills
3. Memory and State



Router

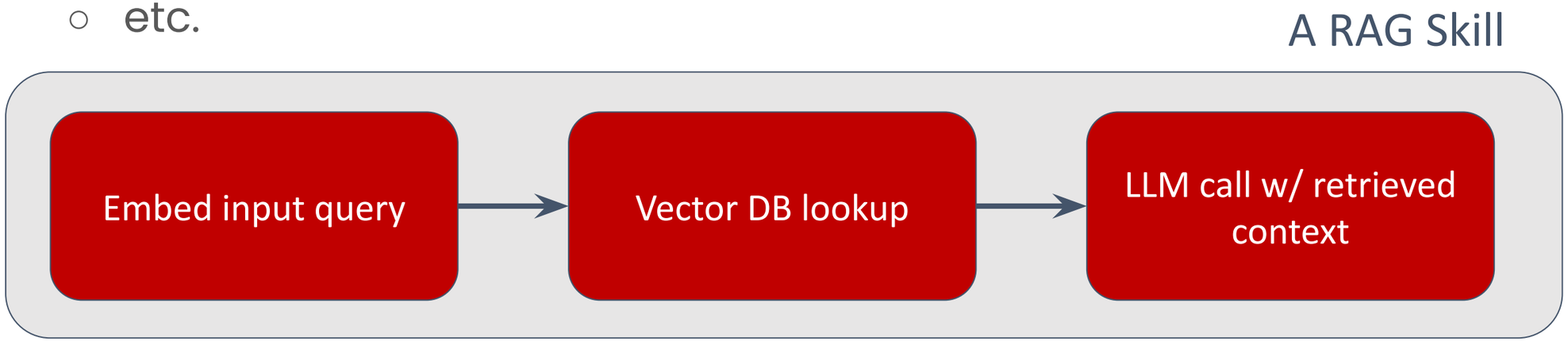
- The main planner of the agent
- Decides which skill or function to call to answer the user's question
- Can take the form of an LLM, an NLP classifier, or even rules-based code
- Some agents distribute router logic throughout the agent, instead of having a single router unit



Skills

A skill is a chain of individual logic blocks that can complete a task

- Each agent will have one or more skills
- Made up of individual steps
  - LLM calls
  - Application code
  - API calls
  - etc.



Memory and State

- Shared state that can be accessed by each component in the agent
- Used to store:
  - Retrieved context
  - Configuration variables
  - Previous agent execution steps

Many LLM APIs rely on receiving each agent step to decide on the next step. Your list of previous steps is often stored in memory.

```
messages = []
messages.append({"role": "system", "content":
"You are a helpful customer support assistant.
Use the supplied tools to assist the user."})
messages.append({"role": "user", "content":
"Hi, can you tell me the delivery date for my
order?"})
messages.append({"role": "assistant",
"content": "Hi there! I can help with that.
Can you please provide your order ID?"})
messages.append({"role": "user", "content": "i
think it is order_12345"})
```

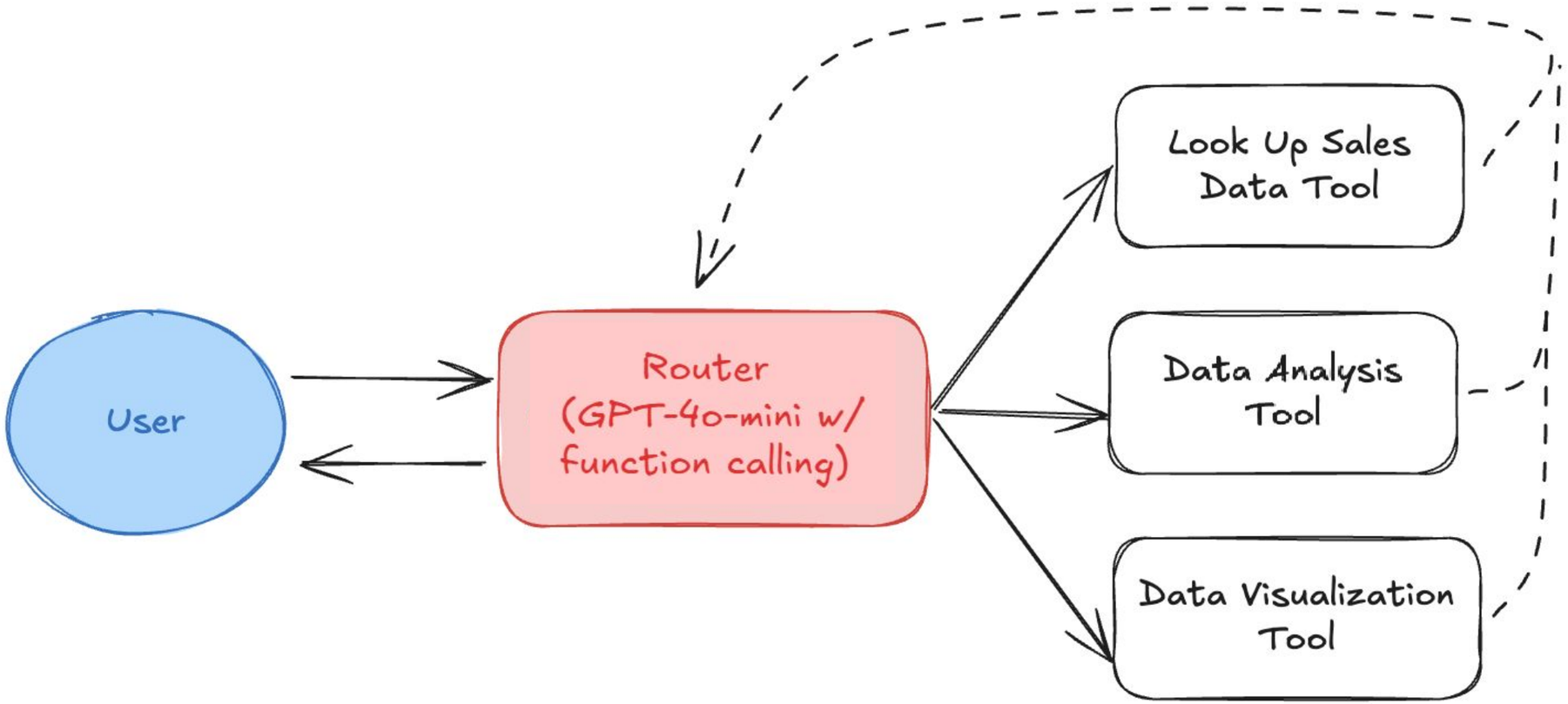
Your Example Agent

Use case: a data analysis assistant that can help you understand sales data from each of your stores

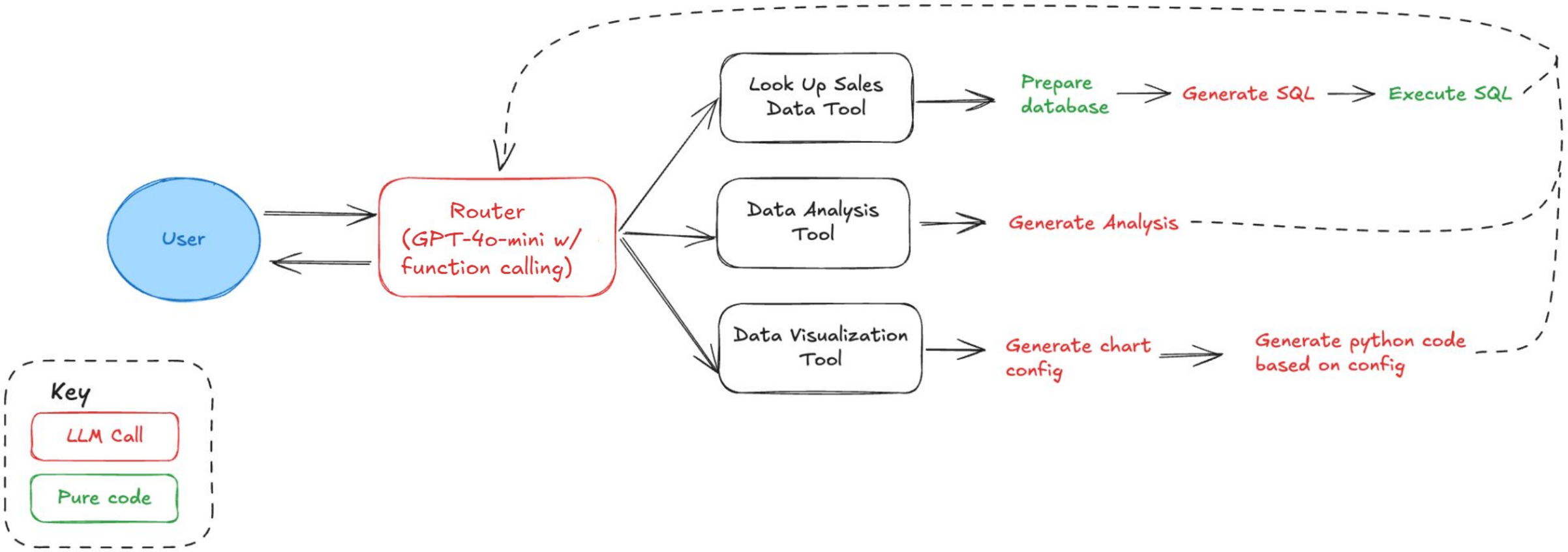
Skills:

- A data lookup skill to query from your attached database
- A data analysis skill to draw conclusions from your data
- A data visualization skill to generate graphs and visualizations about your data

Your Example Agent



Your Example Agent



What you've learned:

1. The major components of agents: Routers, Skills, Memory
2. Your example agent